

A Paintable, Scalable, and Durable Zwitterionic Hydrogel Coating for Enhanced Marine Antifouling Applications

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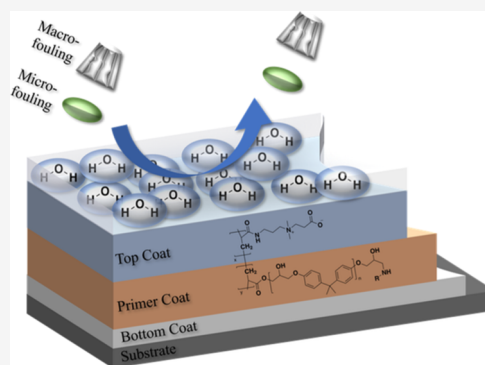


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ABSTRACT: Marine biofouling has been a severe challenge since the increase of maritime trade, significantly impacting the efficiency of ships by increasing drag, fuel consumption, hull corrosion, and even problems related to navigational safety and biological invasions. Commercial antifouling coatings have been developed for many years, but a satisfactory solution has yet to be found due to problems, such as high toxicity, environmental pollution, or high costs. Zwitterionic materials, with their superhydrophilic properties, demonstrate excellent resistance to nonspecific adhesion alongside good biocompatibility, making them promising candidates for marine antifouling applications. However, their superhydrophilic nature makes it difficult to anchor onto hydrophobic substrates, limiting their use. In this study, we presented a paintable, scalable, and durable antifouling coating system made by zwitterionic hydrogel (PSDA-Z), which was covalently attached to substrates through an acrylated epoxy resin primer coat and maintained antifouling performance even after 3 months of high-speed water shearing, high-pressure sandpaper abrasion, and sharp scratching. This PSDA-Z could also easily be applied on various substrates without specific treatments, including epoxy resin, poly(vinyl chloride) (PVC), polyurethane (PU), and wood. More importantly, this coating system achieved excellent antifouling performance comparable to self-polishing coatings (SPCs), the current industry standard in marine antifouling coating, in the Atlantic Ocean field tests for 3 months, suggesting its promise as an effective and ecofriendly alternative for marine antifouling applications.



INTRODUCTION

Since the Age of Discovery, marine transportation has been an essential method of global communication and trade. With globalization, maritime transport has become even more critical due to its large carrying capacity and low cost. However, biofouling, a major challenge in marine transport, significantly decrease economic efficiency and pose environmental risks. Organisms, such as algae and shellfish, adhere to ship hulls, increasing sailing drag, resulting in lower fuel efficiency and higher greenhouse gas emissions.^{1,2} Additionally, fouling can cause hull corrosion³ and pipeline block⁴ and may lead to mechanical failures during navigation. If biofouling organisms are released into non-native marine environments, they may cause biological invasions, threatening local species and ecosystems.⁵

Antifouling coatings offer a cost-effective solution to address the biofouling challenge.^{6–8} Over the years, various commercial products have been introduced, and self-polishing coatings (SPCs) constitute a significant category.^{9,10} The SPCs contain metal-based compounds,¹¹ such as Sn(IV),¹² Cu(II),¹⁰ and Zn(II),¹³ and the coating gradually polishes over time, resulting in the release of biocides, which can prevent the

attachment of marine organisms.¹⁴ However, due to the continuous release of biocides into the aqueous environment, there are significant environmental concerns related to these antifouling coatings. Especially the tributyltin (TBT) contained SPCs, which was the gold standard in the last century and has been used broadly since the 1960s.¹⁵ The TBT can significantly prevent the attaching and growth of the micro- and macroorganisms by affecting their metabolic activity and cell division.¹⁶ However, the TBT is very toxic to marine organisms,¹⁷ with particularly severe effects on the reproduction of shellfish.¹⁸ Even worse, TBT can hardly be degraded in the environment, and its toxicity can be accumulated and transmitted through the food chain, impacting the health of other marine species and threatening public food safety.¹⁸ Therefore, the TBT has been prohibited worldwide since

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2008,¹¹ and copper-based SPCs, which are safer and less toxic, dominate the market.¹⁹ However, the released copper compounds are still a concern, because of the polishing mechanism. To protect the environment, some regions seek to impose legal restrictions or even bans on using copper-based SPCs.^{20–22} Besides the SPCs, fouling release coatings (FRCs), which use physical methods rather than biocides to prevent fouling, offer an environmentally friendly alternative.^{23,24} These coatings have smooth surfaces made of low surface energy materials,^{24,25} such as silicone-²⁴ and fluoro-²⁶ based polymers, reducing the attachment strength of fouling organisms and allowing them to be washed off by water flow. However, FRCs perform poorly at low speeds or in static water, where insufficient water flow leads to fouling accumulation.^{14,24} Additionally, the high cost and maintenance requirements of FRCs limit their broader application.^{27,28}

In addition to commercial antifouling coating products, numerous strategies have been developed in laboratories to address biofouling issues. Environmentally friendly biocides, such as E-conea,^{29,30} indole derivatives,³¹ etc., have been incorporated into SPC systems. Compared to conventional metal-based biocides, these novel compounds effectively target fouling species while minimizing the impact on nontarget organisms. Furthermore, their biodegradability minimizes potential environmental hazards and reduces the risk of toxicity accumulation in the food chain. However, with the polishing over time, the coating thickness decreases, which corresponds to the declines in their antifouling performance.^{32–34} To address this issue, biocides have been chemically modified on the surface, aiming for longer-term antifouling efficacy.^{35–37} Nevertheless, even if they are not released, biocides can still be degraded through oxidation and other chemical reactions in complex marine environments. Additionally, microbial debris can accumulate on the coating surface, leading to a failure in antifouling effectiveness.³⁸

Beyond biocidal approaches, some novel hydrophobic and/or hydrophilic coatings were developed to achieve antifouling purposes in the laboratory.^{39–41} Polydopamine (PDA), a widely studied material, is often employed as an adhesive interlayer between the substrate and the functional top coat due to its self-polymerizing properties and strong intermolecular interactions.^{41–43} PDA provides reliable adhesion to a variety of substrates and offers multiple reactive sites connecting diverse top coat materials. However, the long-term durability of PDA coatings in actual marine environments is still a concern due to their hydrophilic nature, which leads to dissolution tendency, and potential degradation over time.^{44,45} These limitations hinder the widespread adoption of PDA in large-scale marine antifouling applications, particularly in cases where durability and stability are critical. Plasma treatment is another popular surface modification technique that introduces functional groups such as hydroxyl radicals and peroxides to the substrate's surface.^{46–48} This allows polymers to be grafted onto the substrate through interfacial reactions. Although plasma treatment offers an efficient method for surface modification,⁴⁹ scalability of this technique to large applications, such as on yachts or cargo ships, remains challenging.⁴⁴ Additionally, the durability of the grafted layer in harsh marine environments is a significant concern, especially for long-term use.^{44,50} These drawbacks highlight a gap in the practical application of hydrogel-based antifouling coatings, as existing strategies struggle to meet the demands of real-world marine environments.

Zwitterionic materials, featured by their balanced positive and negative charges, have attracted increasing attention these years.⁵¹ Their superhydrophilic nature arises from a dense hydration layer on the surface, distinguishing them from conventional hydrophilic materials such as poly(ethylene glycol) (PEG), which achieve hydrophilicity through hydrogen bonding.¹⁴ This hydration layer allows zwitterions to effectively repel various contaminants, from proteins to microbes and even larger fouling organisms.^{52,53} Notably, zwitterionic materials are biocompatible and nontoxic, making them safe for marine applications, as demonstrated by their successful use in implantable medical devices and drug delivery systems.^{14,54,55} However, applying zwitterionic materials in marine antifouling solutions presents significant challenges. Their superhydrophilic and highly charged nature restricts their solubility to primarily water, complicating production and processing. Additionally, the huge disparity between conventional hydrophobic substrates and superhydrophilic zwitterionic materials makes it difficult to achieve durable adhesion between the substrate and the coating. Furthermore, the strong affinity of zwitterions for water drives the superhydrophilic coating to dissolve or delaminate from the substrate in aquatic environments.^{53,56}

As a result, developing a universally durable and easily applicable coating strategy is essential for successfully using zwitterionic materials in marine antifouling applications. Several methods have been explored to apply zwitterionic materials to hydrophobic substrates, including copolymerization,^{57,58} grafting,^{59,60} and plasma treatment.^{47,61} However, most of these approaches face scalability limitations such as complex synthesis processes, strict application conditions, and difficulties in fabrication. In 2018, we published a highly durable zwitterionic hydrogel tapping method that achieved strong adhesion to various substrates when applied with superglue.⁵⁶ The hydrogel coating maintained nearly zero microbial adhesion over a 3 month coculture with concentrated *Escherichia coli* suspension, showing remarkable antifouling capability and durability of this coating approach. However, the superglue's unpleasant odor and the stamping method hindered its scalable application.

In this study, we presented a novel zwitterionic hydrogel coating strategy, exhibiting the paintability, scalability, and durability required in a real-world coating application while maintaining an excellent antifouling performance (PSDA-Z). The coating can be conveniently obtained by applying a zwitterionic solution to a modified commercial epoxy resin and initiating with UV light. This coating can be easily applied to various substrates, including stainless steel, poly(vinyl chloride) (PVC), polyurethane (PU), and wood, while providing outstanding antifouling performance. The PSDA-Z coating demonstrated high durability in rigorous tests, including 3 months of high-speed water shearing, high-pressure abrasion (490 kPa), and sharp scratching without losing functionality. During a 2 week marine microbial contamination test, the PSDA-Z coating exhibited antifouling performance comparable to copper-based SPCs, the gold standard in marine antifouling, and significantly outperformed commercial epoxy resin and FRC controls. Furthermore, the scaled-up PSDA-Z coating maintained its outstanding antifouling properties in a 3 month field test in the Atlantic Ocean, matching the performance of SPCs while preserving a smooth, glossy, uncontaminated surface. These findings highlight the PSDA-Z coating's potential as an environmentally friendly, paintable,

scalable, and durable solution for sustainable, long-term marine antifouling applications.

EXPERIMENTAL SECTION

Materials. The carboxybetaine acrylamide (CBAA) monomer was synthesized according to a previously published method.⁵⁶ Bisphenol A diglycidyl ether (BPADE), tetrabutylammonium bromide (TBAB), hydroquinone, *N,N'*-methylenebis(acrylamide) (MBAA), Irgacure 2959 (I-2959), acrylic acid (AA), and phosphate-buffered saline (PBS) tablet were obtained from Sigma-Aldrich. The deuterated toluene (toluene-*d*₈) was produced by Cambridge Isotope Laboratories, Inc. The Ulva, *Navicula*, *Cyclotella*, seawater, alga-gro freshwater medium, and alga-gro seawater medium were purchased from the Carolina Biological Supply Company. Stainless steel plates, poly(vinyl chloride) (PVC) plates, and wood plates were obtained from McMaster-Carr. The epoxy resin was Penetrating Epoxy Sealer produced by TotalBoat. Thg (FRC) was Underwater Foul Release Propeller & Running Gear Coating (silane-based fouling release coating) produced by PropGlide. The self-polishing coating (SPC) was Micro CSC (copper oxide-based self-polishing coating) produced by Interlux. The nuclear magnetic resonance (NMR) is the MR 400 produced by Agilent. The high-resolution mass spectrum (HRMS) is characterized by the Thermo Orbitrap Exploris 120 ESI Positive Mode and operated by Dr. Nicholas Peraino in the Lumigen Instrument Center at Wayne State University. The Fourier transform infrared (FT-IR) spectrometer is the Invenio R model produced by Bruker.

Synthesis of Monosubstituted Bisphenol A Epoxy Monomer (2-Hydroxy-3-(4-(2-(4-(oxiran-2-ylmethoxy)phenyl)propan-2-yl)phenoxy)propyl acrylate). 0.49 g of BPADE was added to a 50 mL round-bottom flask containing 10 mL of toluene, followed by 130 mg of TABA and 50 mg of hydroquinone. They were stirred for 15 min at room temperature (rt) to completely dissolve before 0.09 g of AA was added dropwise. The solution was degassed by nitrogen and heated to 70 °C for 4 h. After the reaction, the system was allowed to cool to room temperature before evaporation, and the residue was purified by silica column spectroscopy with hexane: ethyl acetate (v/v) = 3:1. ¹H NMR (400 MHz, toluene-*d*₈) δ 6.81–6.58 (m, 4H), 6.23 (dd, *J* = 17.3, 1.6 Hz, 1H), 5.89 (dd, *J* = 17.3, 10.4 Hz, 1H), 5.26 (dd, *J* = 10.4, 1.6 Hz, 1H), 4.20–4.08 (m, 2H), 3.97–3.82 (m, 1H), 3.75–3.58 (m, 4H), 3.50 (dd, *J* = 11.0, 5.6 Hz, 1H), 2.89 (dq, *J* = 6.0, 3.1 Hz, 1H), 2.24 (ddd, *J* = 46.9, 5.3, 3.3 Hz, 2H), 1.57 (s, 6H). HRMS (ESI): [M + Na]⁺ calculated: (C₂₄H₂₈O₆Na): 435.1778; found: 435.1771.

Synthesis of Acrylate TotalBoat Epoxy Resin (AA-Modified TBER). Ten milliliters of TotalBoat penetrating epoxy resin was placed into a 50 mL round-bottom flask, followed by 2 mg of hydroquinone as the inhibitor and 10 mg of TBAB as the catalyst. The system was stirred for 15 min to dissolve completely, and then 200 μ L of AA (~2 wt % to epoxy resin) was added to the solution dropwise. The mixture was degassed by nitrogen and heated to 70 °C for 4 h. The product was used directly without further purification.

Fabrication of PSDA-Z Coating. The stainless steel plates were sequentially washed with water, ethanol, and acetone and then sanded with 120-grit sandpaper. After surface preparation, the stainless steel plates were coated with Interlux VC performance, which has better adhesive capabilities on metal substrates following its manufacturing protocol as the bottom coat.

After completely drying, the bottom layer was washed and sanded again as the last step to improve the interaction between the bottom coat and the acrylate epoxy resin primer layer. The prepared AA-modified TBER was mixed with the TotalBoat penetrating epoxy hardener (primer:hardener = 2:1 v/v, based on the manual) and applied on the V_c performance surface (100 μ L/cm²). The AA-modified TBER primer coat was semicured in 1–2 h at rt before applying the zwitterionic hydrogel top coat (50 μ L/cm²).

For the zwitterionic top coat solution, 5 g of CBAA monomer was dissolved in 5 mL of water with 50 mg of MBAA (1 wt % to CBAA monomer) and 1.5 mL of AA (30 wt % to CBAA monomer, for

antisediment purpose).⁶² 1 mL portion of 2-propanol (10% of total volume) was added as the surfactant.

Prior to top coat application, 15 mg of I-2959 (0.3 wt % to CBAA monomer) was incorporated to produce a transparent pregel solution. The pregel solution was brushed/sprayed on the AA-modified TBER primer layer and initiated by a UV lamp (365 nm, 100W) for 30 min for a complete reaction. The coated plates were left at rt for an additional 4-day curing period for the AA-modified TBER primer coat per the manufacturer's instructions prior to testing.

Other substrates, including PVC, PU, and wood, were coated using similar procedures but without V_c performance as the bottom coat.

The thickness of each layer was measured by using a digital vernier caliper on the cross section and subsequently calculated. The coating thickness is reported as the mean $\pm$ standard deviation based on three replicate samples (*n* = 3).

Contact Angle Test. The water contact angle on the TBER, AA-modified TBER primer coat, and PSDA-Z top coat was performed using a KSV contact angle instrument equipped with a camera under rt and ambient humidity. A 1 μ L droplet of water was carefully placed on the sample surface and imaged. The water contact angle was then calculated by using ImageJ software.

Antifouling Evaluation. The adhesion of *E. coli* on sample surfaces was used to evaluate antifouling properties. Samples (1 cm $\times$ 1 cm) were placed in a 12-well plate. A 2 mL *E. coli* suspension (OD₆₀₀ = 0.8) was added, and the plates were shaken at 200 rpm at 37 °C for 3 days. After incubation, the microbial-treated samples were transferred to a new 12-well plate and washed three times with PBS solution (filtered through a 200 μ m membrane) to remove loosely attached microbes. The samples were then fixed using 2.5% glutaraldehyde and 2% paraformaldehyde, followed by dehydration in a graded ethanol series and vacuum drying, as per a previously published protocol.⁵⁶ The dried samples were coated with nanogold (SEEVac Conductive IV sputter coater) to enhance the interface conductivity before the scanning electron microscopy (SEM) test (JSM-6510LV SEM). Each sample group consisted of three replicates (*n* = 3). For each sample, microbial adhesion was quantified by averaging the counts from five randomly selected areas (100 μ m² each).

Durability Test. The PSDA-Z-coated sample (stainless steel substrate) was secured in a beaker (*D* = 8.5 cm) containing a 3.5 cm stirring bar and immersed in seawater. The stirring speed was set at 1500 rpm, and the seawater continuously sheared the sample for 90 days at rt. The beaker was sealed with a polyvinyl film to prevent evaporation. Antifouling performance was assessed after this durability challenge and compared with that of freshly prepared samples.

Mechanical Damage Test. For the abrasion test, 400-grit sandpaper (Dura-Gold) was attached to the bottom of a 5 kg weight, while the PSDA-Z sample was secured on a flat surface. The weight was freely placed on the PSDA-Z sample and moved back and forth across the surface; each pull and push (~1 cm) constituted one cycle. Samples underwent 20 cycles of abrasion under applied pressure. For the knife-scratch test, each PSDA-Z coating sample was randomly cut 10 times by using a sharp scalpel. After these mechanical damage challenges, water wettability and antifouling performance were evaluated and compared with those of the freshly prepared samples.

Marine Microbial Adhesion Test. Bare epoxy, SPC, FRC, and PSDA-Z-coated samples (1 cm $\times$ 1 cm) were equilibrated in filtered seawater or DI water for 3 days at room temperature to remove any unreacted residues. *Ulva* was cultured in Alga-Gro Seawater medium for 3 days at room temperature to produce zoospores, which were then diluted to a concentration of 4 $\times$ 10⁴/mL for testing. *Navicula* suspension was diluted in Alga-Gro Freshwater medium to 5 $\times$ 10⁴/mL, while the *Cyclotella* suspension was diluted in Alga-Gro Seawater medium to 6 $\times$ 10⁴/mL and used directly. Coating samples were placed in 12-well plates, with each well containing 2 mL of either *Ulva* zoospore, *Navicula*, or *Cyclotella* suspension. The plates were shaken at 200 rpm at room temperature for both 24 h and 2 weeks, with a 12 h daily exposure to 80 W plant lights. Additionally, 0.5 mL of medium was added every 3 days. After treatment with microbial suspensions, samples were transferred to new 12-well plates and washed three

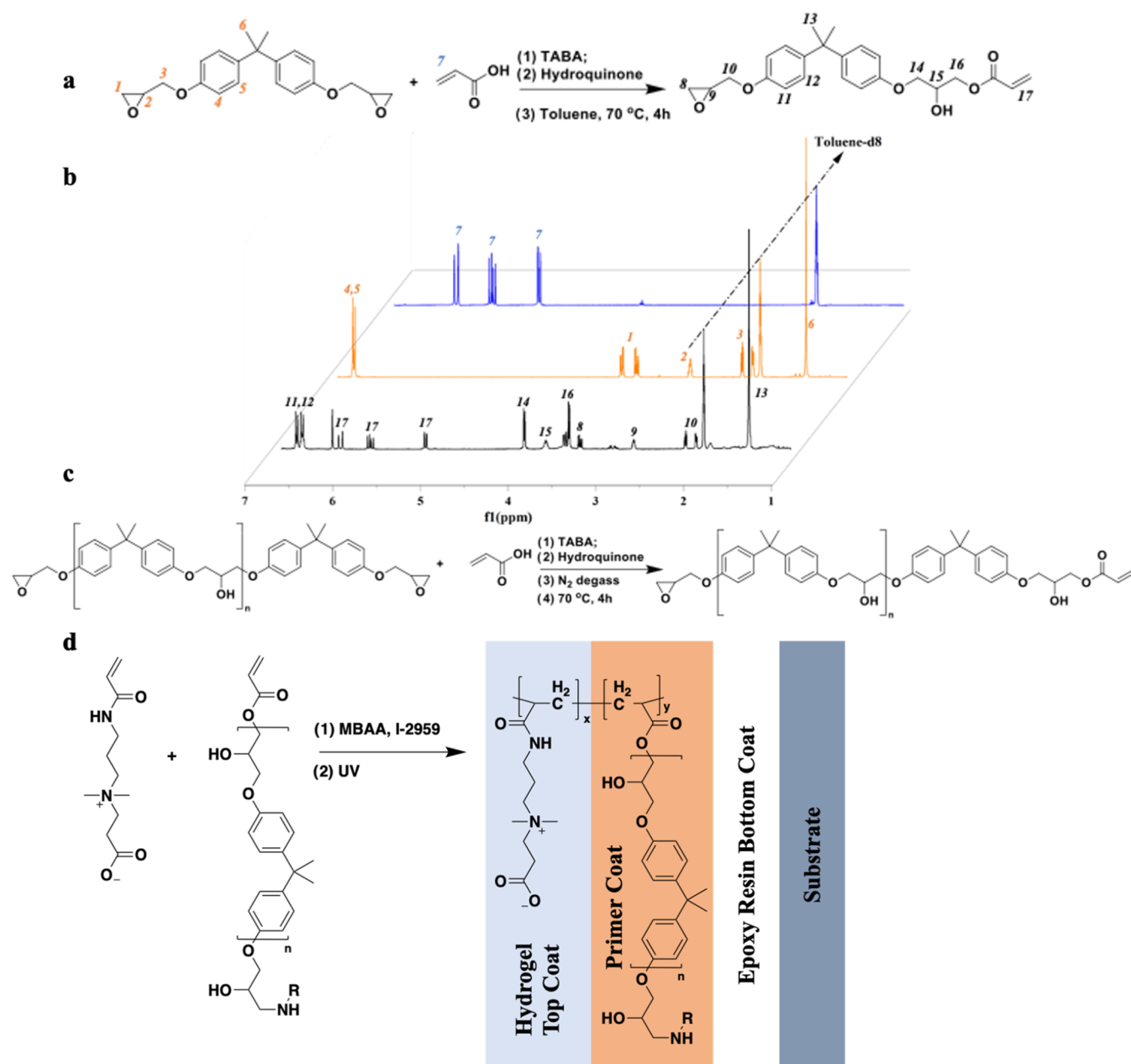


Figure 1. Synthesis and characterization of acrylate epoxy resin. (a) The reaction scheme and conditions of BPADE and AA. (b) NMR spectra of BPADE, AA, and acrylate BPADE with corresponding assignments. (c) The reaction scheme and conditions of BPA-based epoxy resin and AA. (d) The reaction scheme and condition of fabricating PSDA-Z top coat (polymerization reaction in hydrogel top coat and at the top coat/primer coat interface).

times with a PBS solution (filtered through a 0.22 μm membrane) to remove loosely attached microbes. Samples were then fixed with 2.5% glutaraldehyde and 2% paraformaldehyde, dehydrated through a graded ethanol series, and vacuum-dried according to a previously published method.⁵⁶ The dried samples were coated with a nanogold layer using a SEEVac Conductive IV sputter coater before SEM analysis (JSM-6510LV SEM). Each group consisted of three replicate samples, with microbial adhesion quantified by counting microbes across five randomly selected areas (900 μm^2 each) and averaging the counts.

Field Test. A stainless steel panel (15 cm $\times$ 15 cm) was prepared by sanding with 120-grit sandpaper and washing sequentially with water, ethanol, and acetone. After air drying, the panel was coated twice on both sides with Interlux VC Performance epoxy as a bottom coat, following the manufacturer's instructions. The epoxy surface was sanded with 120-grit sandpaper before the new coat was applied. The

PSDA-Z hydrogel coating was applied, as previously described. For comparison, SPC and FRC coatings were prepared as commercial control samples according to their respective manuals.

The prepared panels were submerged in the Atlantic Ocean at Cape Charles, Virginia (N 37.2636, W 76.0158). Panels were positioned approximately 0.5 m underwater by securing them to a floating platform to minimize the tidal effects.

RESULTS AND DISCUSSION

Preparation of Acrylate Epoxy Resin. Epoxy resins are a class of polymer materials known for their excellent adhesion, waterproofing, corrosion resistance, and robust mechanical properties.^{63–65} Due to these attributes, they are widely employed as protective coatings,^{66,67} sealants,^{68,69} and flooring^{70,71} applications. Among the various types of epoxy resins,

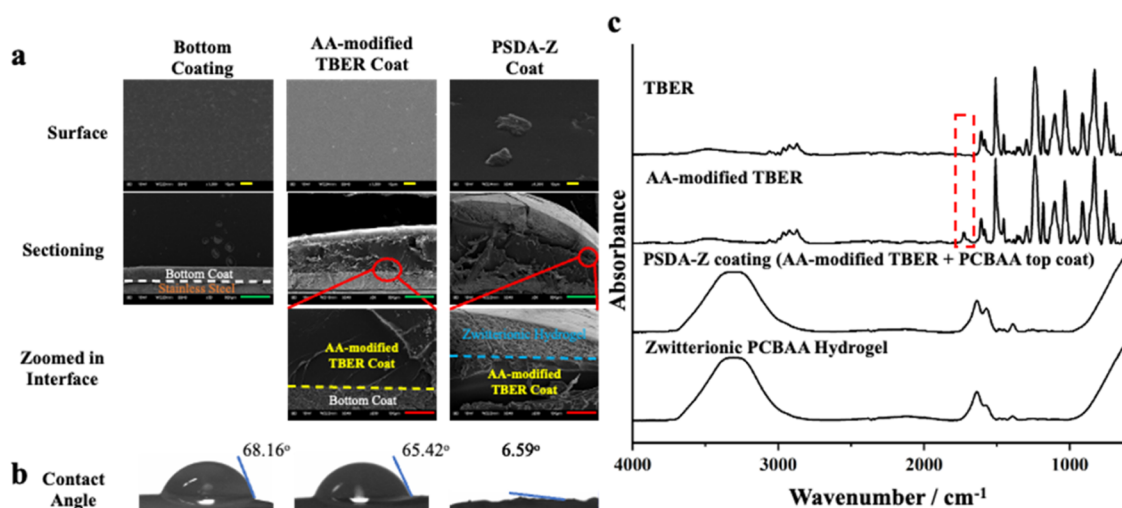


Figure 2. (a) Representative SEM images of the surface (scale bar = 10 μm), cross section (scale bar = 500 μm), and locally zoomed-in (scale bar = 100 μm) for commercial epoxy resin, AA-modified TBER, and the PSDA-Z coating system containing both the primer and the top coat. (b) Water contact angle of TBER (unmodified epoxy resin), AA-modified TBER, and the PSDA-Z top coat surface. (c) FT-IR spectra of the TBER, AA-modified TBER, PSDA-Z top coat surface, and bare PCBA hydrogel surface.

bisphenol A (BPA)-based epoxy resin is the most used,⁷² owing to its cost efficiency and reliable performance.⁷³ BPA-based epoxy resin can be easily cured by mixing it with polyamines under mild conditions. Many modifications have been applied to this structure, and acrylate modification is particularly valuable, as it introduces an alkene group into the polymer's main structure, providing a secondary functional group.⁷⁴ In this bifunctional structure, the high molecular weight of the epoxy resin enables strong immobilization on various substrates, such as metal and plastic, via strong intermolecular interactions and π - π stacking. The introduced alkene groups do not react with curing agents but can react with other alkene-based molecules. Compared to a modification on the curing agent,⁷⁵ we propose that modification on the epoxy resins offers superior durability. The extensive entanglement and twisting of long epoxy polymer chains enhance intermolecular interactions, complementing primary covalent bonds that may degrade over time in harsh environments, thereby providing sustained stability in the epoxy resin-modified system.

We first evaluated the modification chemistry by reacting acrylic acid (AA) with bisphenol A diglycidyl ether (BPADE), the monomer for BPA-based epoxy resin polymers (Figure 1a). The amount of AA was carefully controlled to ensure that only one epoxy functional group was consumed. Tetrabutylammonium bromide (TABAB) was used as the catalyst, and hydroquinone was added to minimize unwanted polymerization due to heating.^{76–78} The product was purified and characterized using NMR (Figure 1b) and high-resolution mass spectrometry (HRMS) (Figure S1). As anticipated, the monoacrylated BPADE was successfully obtained, confirming the feasibility of the modification reaction. This reaction condition was subsequently applied to a commercial epoxy resin (Figure 1c), TotalBoat Epoxy Resin (TBER), and the resulting product; partially acrylated TotalBoat Epoxy Resin (AA-modified TBER), was then mixed with the recommended commercial hardener, following the manufacturer's guidelines. This mixture was directly applied to the substrate as a primer without any additional treatment.

Fabrication of PSDA-Z Coatings. To enhance the adhesion strength on metal substrates, Interlux VC Performance was applied as a bottom coat on stainless steel (this step was not required for plastic and wood substrates). Subsequently, the primer coat (AA-modified TBER) and the superhydrophilic zwitterionic hydrogel top coat were applied, as shown in Figure 1d. The bottom coat and AA-modified TBER primer coat were prepared according to their respective commercial guidelines, and the pregel solution for the top coat was applied while the AA-modified TBER primer coat was in a semicured state to facilitate the effective migration of acrylate groups to the primer coat–top coat interface. Gelation was initiated with a 365 nm UV lamp, enabling the formation of the zwitterionic hydrogel top coat and the covalent binding to the primer coat through free-radical polymerization of alkene functional groups at the interface, thus creating the PSDA-Z coating system. The samples were then left at rt for 4 days to fully cure the epoxy-based primer layer, following the commercial guidelines, and subsequently equilibrated in deionized water or filtered seawater for at least 24 h to hydrate the superhydrophilic zwitterionic top coat prior to further testing.

Optimization of the AA-Modified Primer Coat. The reaction condition for acrylating BPADE was applied to the modification of TBER, and the primer formulation was refined by adjusting the amount of AA added to the commercial epoxy resin for acrylate modification. The durability of the resulting formulation was evaluated under both static and dynamic water environments (Table S1). In dynamic seawater conditions, the 8 wt % AA-modification resulted in the detachment of the primer coat from the substrate within 2 weeks, while the 4 wt % AA-modification showed pinpoint damage to the primer coat (Table S2). After 1 month, edge damage expanded in the 4 wt % AA-modification sample, and larger detachment occurred in the 6 wt % AA-modification sample. By day 75, extensive detachment was noted in both the 6 wt % and central regions of the 4 wt % AA-modification samples, whereas the 2 wt % AA-modification remained intact throughout the test. A similar tendency was observed under static seawater conditions (Table S3). We hypothesized that an

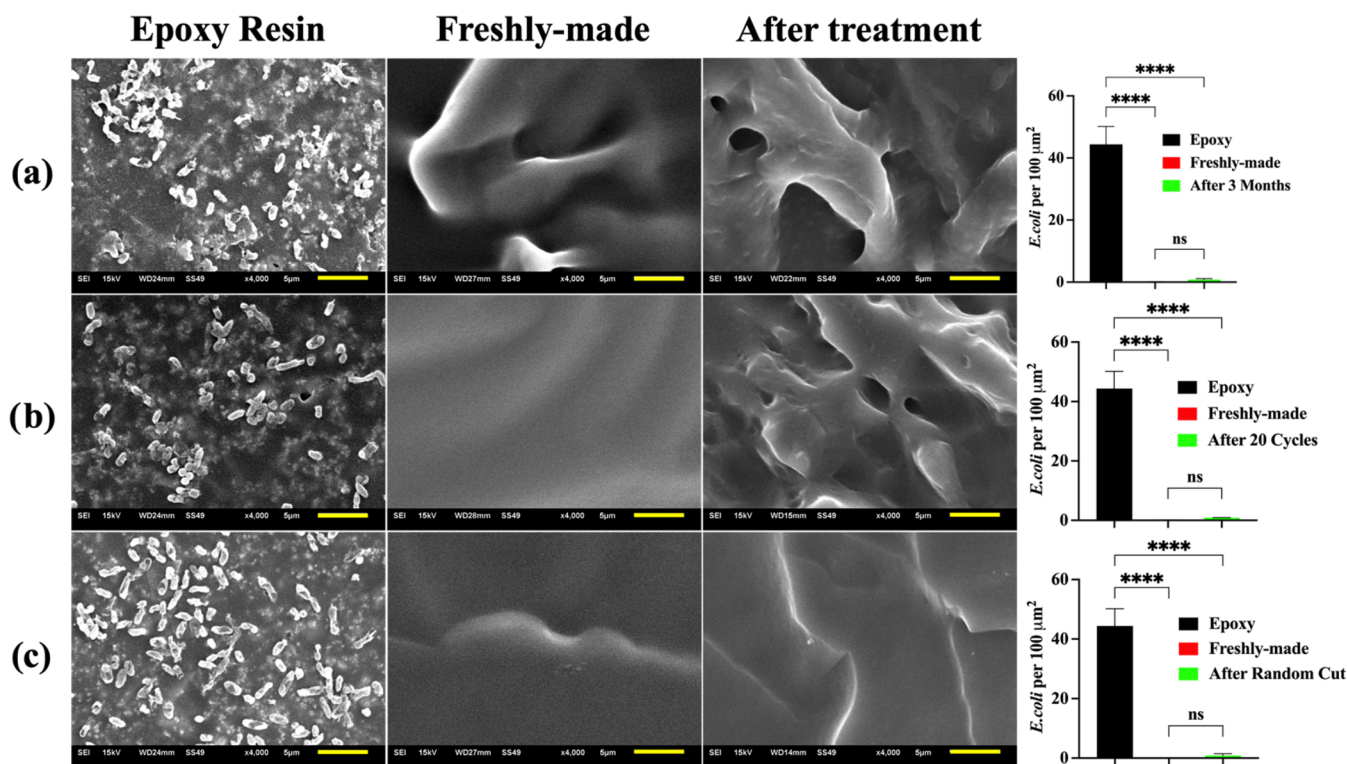


Figure 3. Durability and mechanical damage tests of PSDA-Z coating samples, which were quantitatively assessed by comparing antifouling properties before and after each treatment. (a) Treatment: high-speed (1500 rpm) seawater shearing for 90 days at room temperature. (b) Treatment: abrasion using 400-grit sandpaper for 20 cycles under a 490 kPa pressure. (c) Treatment: random scratching conducted by making 10 cuts with a scalpel. The antifouling capability was evaluated by measuring bacterial adhesion (*E. coli*, OD₆₀₀ = 0.8). For each sample, five areas were imaged by SEM and averaged; data were presented as mean $\pm$ standard deviation ($n = 3$). Statistical analysis was performed using one-way analysis of variance (ANOVA) with multicomparison. ns: $p > 0.05$, ****: $p < 0.0001$. Scale bar = 5 μ m.

excess of AA consumes additional epoxy groups, thereby reducing cross-linking between epoxy resin polymer chains and weakening the adhesion properties. Additionally, the zwitterionic hydrogel top coat, despite being covalently bonded to the primer coat, tended to dissolve in water, contributing to the primer coat's detachment from the substrate. Therefore, the 2 wt % AA-modification to epoxy resins was selected for further evaluation due to its superior durability compared to other formulations.

Characterization of PSDA-Z Coatings. The PSDA-Z coating was characterized by using scanning electron microscopy (SEM) for surface and cross-sectional views (Figure 2a). The Interlux VC Performance bottom coat and the primer coat exhibited smooth and dense surfaces, whereas the PSDA-Z top coat displayed a gelatinous texture. The cross-sectional images provided more precise insights; the distinct differences in density and toughness between the bottom coat and the primer coat were observable. In contrast, the PSDA-Z top coat showed a porous structure due to the superhydrophilicity of the zwitterionic hydrogel, which created voids during the freeze-drying process. Notably, unlike our previously published DURA-Z coating, which featured an interpenetrating layer between the zwitterionic hydrogel and the cured superglue,⁵⁶ the PSDA-Z coating lacked this interpenetration structure because the zwitterionic hydrogel top coat adhered to the epoxy resin layer through covalent bonds.

The thickness of each layer was measured using a digital vernier caliper on the cross section and subsequently calculated. In a typical PSDA-Z coating system, the bottom

coat measured $343 \pm 91 \mu$ m, the primer coat measured $1296 \pm 64 \mu$ m, and the top coat measured $823 \pm 40 \mu$ m in its dry state. After equilibrium in seawater for 24 h, the top coat swelled to $2013 \pm 78 \mu$ m.

Water contact angles were measured for the TotalBoat Epoxy Resin (TBER, unmodified commercial epoxy resin), AA-modified TBER primer coat, and PSDA-Z top coat to assess their hydrophilicity (Figure 2b). The TBER displayed a contact angle of 68° , indicating its hydrophobic nature. After acrylate modification, the water contact angle decreased slightly to 65° , showing minimal changes to its inherent properties. A significantly decreased water contact angle appeared on the PSDA-Z top coat at 6° , highlighting the superhydrophilicity of the zwitterionic hydrogel top coat. A similar ultralow water contact angle was also observed on the PSDA-Z coating system on other substrates, including PVC, PU, and wood (Figure S2), indicating the universal applicability of PSDA-Z on various substrate materials. This extreme hydrophilicity is expected to enhance the coating's antifouling performance in marine applications.

The modified epoxy resin primer coat and hydrogel top coat were further characterized using Fourier transform infrared (FT-IR) spectroscopy (Figure 2c). It is important to note that the mobility of acrylate functional groups was significantly restricted after solidification. Therefore, the TBER and AA-modified TBER samples were tested before curing. The FT-IR spectra of AA-modified TBER were nearly identical to those of TBER, except for an additional peak at approximately 1716 cm^{-1} , corresponding to the C=O stretching of the α,β -unsaturated ester.⁷⁹ This observation, along with our small-

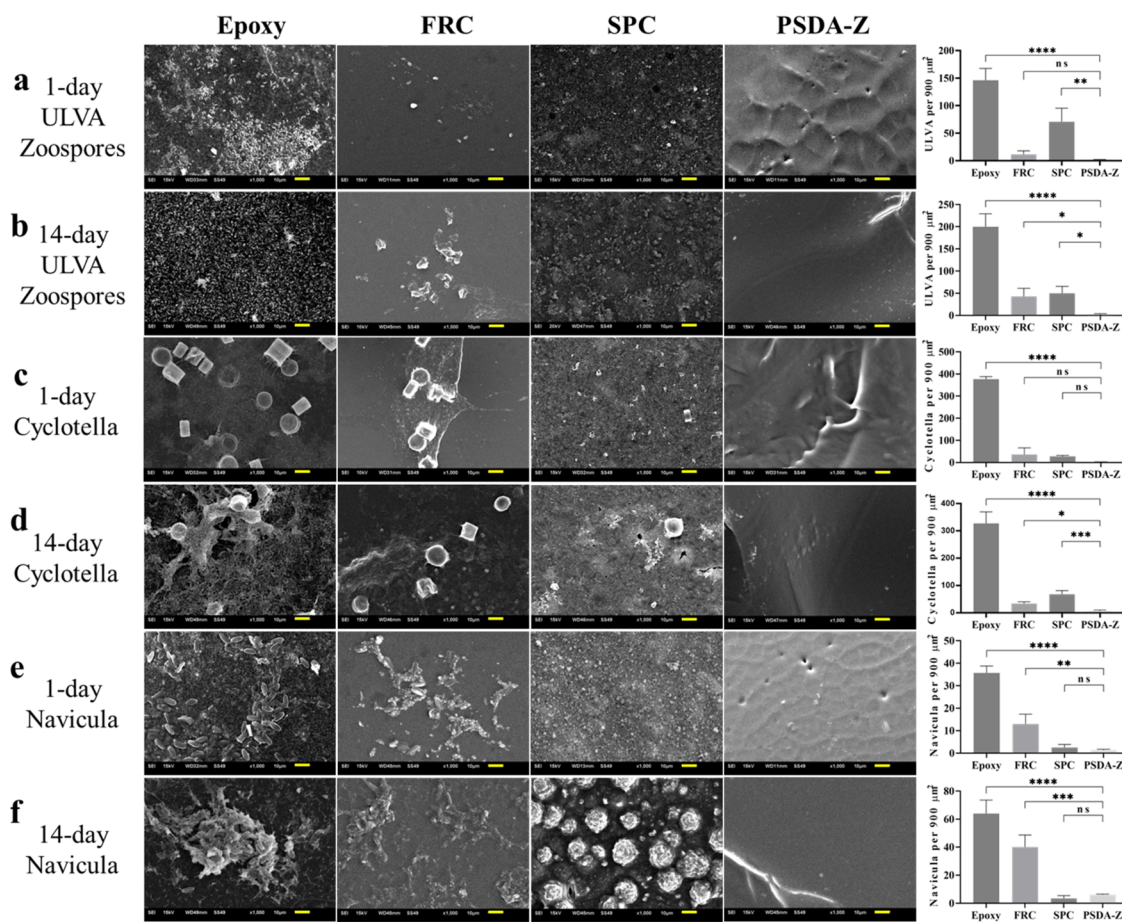


Figure 4. Representative SEM images of commercial epoxy resin, FRC, SPC, and PSDA-Z coatings on stainless steel after coculture with *Ulva* zoospores (marine algae, a, b), *Cyclotella* (marine diatom, c, d), and *Navicula* (freshwater diatom, e, f) for 1 day and 14 days. For each sample, five areas were imaged by SEM, with the number of attached microbes counted and averaged. Data were presented as mean $\pm$ standard deviation ($n = 3$). Statistical analysis: one-way ANOVA with multiple comparisons. ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$. Scale bar = 10 μm (due to the significantly higher microbial counts in the epoxy resin group, this group was excluded when analyzing the difference between FRC, SPC, and PSDA-Z samples using one-way ANOVA).

molecule reaction mechanism, suggested that the acrylate group was successfully incorporated into the BPA epoxy resin polymer chain. Additionally, the FT-IR spectra of the PSDA-Z top coat matched that of the original zwitterionic poly(carboxybetaine acrylamide) (PCBAA) hydrogel, confirming the successful attachment of the superhydrophilic material to the hydrophobic substrate.

Durability and Mechanical Damage Test. Compared with conventional antifouling coatings such as SPCs and FRCs, zwitterionic hydrogels exhibit extreme superhydrophilicity, significantly distinguishing them from metal and plastic substrates. The highly charged structures of zwitterionic materials create a dense hydration layer, providing exceptional antifouling capabilities. However, the high-water solubility of zwitterions results in a strong tendency to dissolve in an aqueous environment. Consequently, long-term durability of zwitterionic coatings (exceeding days or a few weeks) has rarely been reported.^{56,80}

In this study, we evaluated the durability of our PSDA-Z coating by immersing it in seawater for 3 months under high-speed shearing (1500 rpm, ≈ 13 knots on the sample surface). The hydrophilicity and antifouling performance were assessed by measuring the water contact angle (Figure S4) and microbial adhesion (Figure 3a), respectively. As shown in

Figure S4, the water contact angle remained nearly identical to that of the freshly prepared sample even after extended exposure to high-speed water shearing. In addition, the morphology and antifouling capabilities of PSDA-Z coating were slightly affected by this rigorous test (Figure 3a). SEM images showed that the treated sample had a similar gelatinous texture to the freshly prepared samples, with negligible microbial adhesion after 3 days of coculturing with *E. coli*, demonstrating excellent antifouling properties. The robust durability of the PSDA-Z coating system is attributed to the abundant strong covalent bonds between the hydrogel top coat and the AA-modified primer coat. Additionally, the cross-linked structure of the hydrogel further enhances its durability by providing extensive anchoring points across the primer coat–top coat interface.⁸⁰ Compared to traditional brush polymer coatings, where each grafted polymer chain relies on a limited number of individual anchoring points and may detach if these fail, the hydrogel's cross-linked polymer network enables interconnected chains to share anchoring points. This ensures that the hydrogel coating maintains its integrity, coverage, and stability even when some anchoring points are compromised.

The PSDA-Z coating was further evaluated for mechanical durability. After undergoing 20 cycles of abrasion with coarse

sandpaper (400 grit) under high pressure (490 kPa), the coating retained its superhydrophilicity and antifouling capability with a water contact angle of 7.22° and minimal microbial adhesion. To simulate potential scratching scenarios, scalpel-induced scratches were applied to the PSDA-Z top coat. Notably, there was no significant change in the water contact angle between the scratched and freshly prepared samples, indicating a sustained superhydrophilicity. Additionally, almost no *E. coli* adhered to the scratched samples. These results highlight the PSDA-Z coating's long-term durability, exceptional antifouling performance, and high resistance to mechanical damage.

Lab-Based Antifouling Performance. To simulate microbial contamination in an aquatic environment, the PSDA-Z coating samples were exposed to various aquatic microbes, including *Ulva* zoospores (marine green algae), *Cyclotella* (marine diatom), and *Navicula* (freshwater diatom), for both 24 h and 14 day trials in the laboratory, and the antifouling performance was quantified by counting the adhered microbes under SEM (Figure 4). The commercial unmodified epoxy resin and antifouling products, such as silane-based FRCs and copper oxide-containing SPCs, were used as control groups. Figure 4 shows that the unmodified epoxy resin was heavily contaminated by all types of microbes within 24 h, particularly by marine species. The FRC samples exhibited improved performance compared to the unmodified epoxy resin, with significantly fewer attached microbes in the 24 h trial, while increased over the 14 day coculture period. The SPC samples, considered the gold standard in marine antifouling, exhibited the best performance, with a minimal increase in adhered microbes after 14 days compared to 24 h; notably, the number of *Ulva* zoospores even decreased over this period. We hypothesize that biocides released from the SPC initially killed microbes on the surface, but their bodies or debris adhered to the coarse surface during the early hours. Over time, the sustained release of biocides effectively controlled microbial concentrations in the environment, thereby limiting further attachment. Compared to commercial antifouling coatings, our PSDA-Z samples exhibited better antifouling performance against all species across 24 h and 14 day trials. SEM images revealed that few microbes or debris adhered to the surface, even after 2 weeks of coculture. Additionally, after 24 h and 14 day immersion and agitation in freshwater or seawater (Figure 4), the morphology of the PSDA-Z samples showed minimal change compared to that of freshly prepared samples (Figure 3), further demonstrating the coating's durability and efficacy.

Field Test Results. To evaluate the performance of our PSDA-Z coating in a natural aquatic environment, the PSDA-Z coating was applied on stainless steel plates and tested in the Atlantic Ocean. The PSDA-Z coating samples were scaled up to 6 cm $\times$ 10 cm from the lab-based size of 1 cm $\times$ 1 cm and immersed at approximately 0.5 m underwater by hanging them on a floating platform. For comparison, the unmodified epoxy resin, FRC, and SPC samples were tested under the same conditions.

After 1.5 months, the first evaluation was conducted. Samples were gently wiped with gloves, rinsed with seawater, and inspected by following common protocols such as ASTM D6990-05 (Figure 5). The unmodified epoxy resin showed heavy contamination, with roughly 40% of the surface covered by algae and *Bryozoan*. The FRC samples performed better but still exhibited *Bryozoan* attachment. As the gold standard in

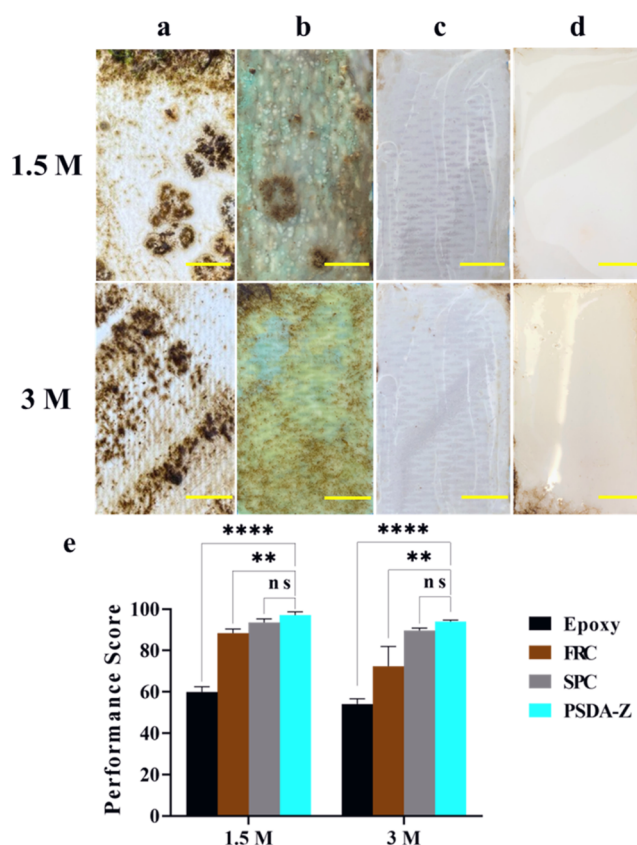


Figure 5. Field test conducted in Cape Charles, Virginia (37.2679° N, 76.0174° W) for 1.5 and 3 months. (a) Commercial epoxy resin coating, (b) FRC, (c) SPC, (d) PSDA-Z coating, and (e) performance scores based on ASTM D3623-78a. The biofouling area on each sample was analyzed using ImageJ, with data presented as mean $\pm$ standard deviation ($n = 3$). Statistical analysis: one-way analysis of variance (ANOVA) with multicomparison. ns: $p > 0.05$, **: $p < 0.01$, ****: $p < 0.0001$. Scale bar = 2 cm.

marine antifouling, SPC samples displayed the best performance among commercial coatings with minimal fouling limited to corners, leaving most of the surface clean. In comparison, the PSDA-Z coating samples matched the performance of SPCs, maintaining a clean, slippery, and glossy surface due to their superhydrophilic properties. After 3 months of marine testing, similar trends were observed. The unmodified epoxy resin and FRC samples showed increased contaminations, resulting in performance scores of approximately 54 and 72, respectively (where a higher score indicates better antifouling performance, based on ASTM D3623-78a) (Figure 5a,b,e). The SPCs (Figure 5c) largely remained clean, with some sediment and algae observed in limited areas, likely due to the coating's coarse texture. Minor algae growth was also found at the edges of the PSDA-Z coating samples (Figure 5d), which we attribute to edge effects. However, nearby growing plants suggest that the PSDA-Z coating is environmentally safe in marine settings.

CONCLUSIONS

In summary, we developed a paintable, scalable, and durable zwitterionic hydrogel-based antifouling coating system composed of a superhydrophilic top coat and a modified hydrophobic epoxy resin primer coat applicable across various substrates. The numerous, robust covalent bonds between the

hydrogel top coat and modified epoxy primer layer provided the PSDA-Z coating with exceptional durability (exceeding 3 months) while maintaining excellent antifouling performance in laboratory tests. The straightforward one-pot synthetic process, combined with the use of readily available commercial epoxy-reinforcement-based coating products, significantly reduces production costs. Moreover, the scalable application process, which requires no specialized equipment, further supports widespread adoption. While initial implementation may involve higher costs for UV curing systems, these are offset by long-term savings in reduced fouling, maintenance, and environmental compliance. By leveraging widely accessible industrial materials and processes, this approach enhances economic feasibility and industrial scalability. This cost-effectiveness, together with its demonstrated antifouling performance during 3 months of field testing in the Atlantic Ocean—comparable to self-polishing copolymers (SPCs), the marine antifouling gold standard—makes the PSDA-Z system a highly promising solution for real-world applications. Furthermore, with its demonstrated biosafety characteristics, the PSDA-Z coating offers significant environmental benefits, contributing to the protection of marine ecosystems.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.4c04595>.

Optimization of epoxy resin modification (Table S1); dynamic test of PSDA-Z durability with different AA-modified TBER formulations (Table S2); static test of PSDA-Z durability with different AA-modified TBER formulations (Table S3); the HRMS of monosubstituted bisphenol A epoxy monomer (2-hydroxy-3-(4-(2-(4-(oxiran-2-ylmethoxy)phenyl)propan-2-yl)phenoxy)-propyl acrylate) (Figure S1); the water contact angle of freshly prepared PSDA-Z coating on PVC, PU, and wood (Figure S2); (a) representative SEM images of the surfaces of commercial epoxy resin and PSDA-Z coating systems on various substrates (stainless steel, PVC, PU, and wood) after 3 days of coculture with *E. coli* suspension (OD₆₀₀ = 0.8); (b) statistical analysis of *E. coli* adhesion for each group (Figure S3); the water contact angles of PSDA-Z coatings that were freshly prepared, subjected to 3 month seawater shearing, abrasion, and scratching challenges (stainless steel substrate) (Figure S4); speed calculation (PDF)

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Notes

The authors declare the following competing financial interest(s): Zhiqiang Cao is a co-founder of Repela Tech LLC.

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